

Luke Brzozowski

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Result-driven with 3+ years work experience in Engineering, seeking positions in Embedded Systems or Software Engineering.

EDUCATION

The University of Michigan - Ann Arbor | *Bachelor of Science in Engineering in Computer Engineering* Dec 2025
Courses: **Computer Architecture** (old EECS 470), Embedded System Design, Embedded Control, Signals and Systems, Digital Logic Design, Computer Security, Computer Vision, Computational Linear Algebra, Data Structures and Algorithms

EXPERIENCE

AgenticFruit/isba.ai, *Technical Co-Founder* | Ann Arbor, MI May 2025 – Present

- Built agentic AI coding system (**TypeScript/LangGraph/Next.js**) with 90% code accuracy and 60% dev time reduction
- Implemented RAG system using Claude, Pinecone, and prompt engineering techniques, achieving 92% user query satisfaction
- Developed AI hotel administration platform with automated full-stack code generation (QloApps/Stripe/ElevenLabs integration)

Robert Bosch, *Systems Engineering Intern* | Plymouth, MI May 2024 – Jan 2025

- Drove full system delivery of **embedded**, voice-activated ADAS system—spanning **C++** dev, **Raspberry Pi** implementation, vehicle integration, global project management, and deployment. Optimized Google Dialogflow (C++) achieving 90% intent classification accuracy and 200ms response. Delivered market-ready system showcased at CES 2025 to 10,000+ attendees.
- Introduced and developed automation scripting solutions using **Python** (Selenium, pandas) and Excel scripting to streamline data collection, processing, and bulk database updates, resulting in 90% reduction in manual labor for associated tasks

UTAC Inc., *Software Development Intern* | Northville Township, MI May 2023 – Aug 2023

- Utilized LabVIEW to enhance software functionality by integrating email, data acquisition systems, and data export features
- Developed live power monitoring and logging system for test stands with cost reporting module for usage analytics

UoM Formula SAE Electric, *Autonomous Team Member* | Ann Arbor, MI Aug 2023 – Jan 2024

- Developed trajectory control system for autonomous vehicle, increasing tracking accuracy by 25% over previous versions

PROJECTS

Piazza Post Categorizer | *Machine Learning (ML), C++, Binary Search Tree*

- Developed ML-based classifier for online forum posts (Piazza), achieving 90% accuracy on labeled datasets

AI Product Catalog To Image Generator | *Typescript, Bash, Google Search API, Gemini API*

- Built product catalogs to studio-grade images AI pipeline with 99% accuracy, negligible hallucinations, and 1s/item throughput

Out-of-Order RISC-V Microprocessor | *SystemVerilog, RISC-V, Python, System Verification*

- Designed 32-bit out-of-order RISC-V processor achieving 12.87ns critical path and 15% IPC improvement over normal 5-stage pipeline. Developed verification framework using **SystemVerilog** testbenches and Python automation
- Constructed modular and configurable GUI visual debugger based on Python (Tkinter) and VCD file parsing

Embedded Virtual Spring System | *C, Firmware, MMIO, Interrupts, Embedded Debugging*

- Developed interrupt-driven **C firmware** for real-time motor control simulating a virtual spring via PWM using MMIO
- Utilized oscilloscopes and logic analyzers for system validation and debugging, automated testing with Python scripts

Accelerometer Data Logger | *C++, Arduino IDE, PCB Design, Microcontroller Programming*

- Designed and built **ESP32**-based PCB for 3-axis acceleration logging and real-time upload to Google Sheets for data acquisition

Motion-Sensed Alarm System | *PIR Sensors, ESP32, IoT, Tinkercad, Rapid Prototyping*

- Developed IoT security system with PIR detection, wireless alerts, and live video streaming, achieving 98% detection accuracy
- Designed waterproof housing with CAD and 3D printing, using iterative prototyping to deliver durable, weather-resistant system

NASA Mars Rover (Prototype) | *C++, Arduino Uno, Arduino IDE, 3D Printing*

- Led team of 4 in prototyping model of NASA Mars Rover within design constraints using rapid prototyping, engineering design standards, and requirements management. Implemented collision avoidance and functional robotic arm systems
- Designed, 3D printed, and tested multi-terrain wheels optimized for stability and traction across various surface conditions

LEADERSHIP

National Gaming Competition

- Founded & led 120-member team in securing 2nd place in three-month global gaming competition among hundreds of teams

TECHNICAL SKILLS

Languages	C, C++, Python, Bash, SQL, TypeScript, JavaScript, Markdown, SystemVerilog, ARMv7-M Assembly, MATLAB
Tech	Node.js, React, CSS, LangGraph, OpenAI, Claude, Supabase API, workflow automation, prompt engineering, OOP
Tools	Git, Docker, Jira, Quartus Prime, Arduino IDE, Verdi, logic analyzer, Simulink, oscilloscope, soldering, 3D printing
Embedded	Raspberry Pi, ESP32, STM32Nucleo, Linux (Arch, Ubuntu), RTL Design, KiCad, CAN bus, I2C, UART